Final Exam - Discrete Mathematics
B. Math II

17 November, 2025

(i) Duration of the exam is 3 hours.

(ii) The maximum number of points you can score in the exam is 100 (total = 105).

(iii) You are not allowed to consult any notes or external sources for the exam.

Name: ____________________________

Roll Number: ____________________________

1. Let $\mu : \mathbb{N} \to \mathbb{N}$ denote the Möbius function.

   (a) (5 points) Prove that

$$\sum_{d|n} \mu(d) = \begin{cases} 1, & \text{if } n = 1, \\ 0, & \text{if } n > 1. \end{cases}$$

   (b) (10 points) With justification, determine $\sum_{n \le x} \mu(n) \lfloor \frac{x}{n} \rfloor$.

Total for Question 1: 15

2. Recall that Bernoulli numbers $\{B_n\}$ are defined by the generating function

$$\frac{t}{e^t - 1} = \sum_{n=0}^{\infty} B_n \frac{t^n}{n!}.$$

   (a) (10 points) Prove the recursive relation

$$\sum_{k=0}^{n} \binom{n+1}{k} B_k = 0, \quad n \ge 1,$$

   with $B_0 = 1$.

(b) (5 points) Show that $B_n = 0$ for all odd $n > 1$.

Total for Question 2: 15

3. (15 points) Let $\mathcal{A} = \{A_1, A_2, \ldots, A_m\}$ be a family of finite sets. Prove that $\mathcal{A}$ has a *system of distinct representatives (SDR)* (i.e., one distinct element can be chosen from each set) if and only if for every subfamily $\{A_{i_1}, \ldots, A_{i_k}\}$, we have

$$\left| A_{i_1} \cup \cdots \cup A_{i_k} \right| \geq k.$$

Total for Question 3: 15

4. (20 points) Let G be a bipartite graph. A *matching* in G is a set of edges with no shared endpoints. A *vertex cover* is a set of vertices that touches every edge in E. Prove that

$$\boxed{\nu(G) = \tau(G)},$$

where $\nu(G)$ is the size of a maximum matching and $\tau(G)$ is the size of a minimum vertex cover.

Total for Question 4: 20

5. Let $n \geq 2$ be an integer. Two Latin squares of order n are said to be *orthogonal* if, when superimposed, every ordered pair of symbols occurs exactly once. A collection of Latin squares is said to be *mutually orthogonal* if every distinct pair in the collection is orthogonal.

   (a) (10 points) Show that the number t of mutually orthogonal Latin squares (MOLS) of order n satisfies
   $$t \leq n - 1.$$

   (b) (10 points) Let q be a prime power. Using the finite field $\mathbb{F}_q$, construct explicitly $q-1$ MOLS of order q (with justification), thereby showing that the upper bound is attained in this case. (Hint: You may define $L_a(i,j) = ai + j \pmod{q}$ for $a \in \mathbb{F}_q^*$.)

Total for Question 5: 20

6. Let $A = [a_{ij}] \in \mathbb{C}^{n \times n}$ be a complex matrix.

   (a) (10 points) Prove that:
   $$|\det(A)| \leq \prod_{i=1}^{n} \|\mathrm{row}_i(A)\|_2,$$
   where $\mathrm{row}_i(A)$ denotes the i-th row of A.

   (b) (10 points) Let every entry of A satisfy $|a_{ij}| \leq 1$. Prove that $|\det(A)| = n^{\frac{n}{2}}$ if and only if $AA^* = nI_n$ and every entry of A is of unit modulus.

Total for Question 6: 20